Hamza Khan

Senior Backend Engineer, Architect & Tech Lead

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SUMMARY

6+ years designing, building, and operating distributed systems and cloud-native backend platforms — from high-availability APIs to AI-powered clinical workflows. Architected and led a HIPAA-compliant telehealth platform from the ground up; experienced in service boundaries, async pipelines, AWS infrastructure, observability and mentoring engineers.

EXPERIENCE

Ignis Health — Senior Software Engineer · Backend / Architect / Tech Lead **Dec 2022 – Present**

USA (Remote)

- Architected and led development of Axis-360, a HIPAA-compliant telehealth platform built from the ground up — covering clinical workflows, scheduling, billing, and AI-assisted documentation.
- Owned backend service architecture and boundaries; designed scalable APIs and data models using Django REST Framework and PostgreSQL for thousands of daily active users.
- Built fault-tolerant async pipelines using Celery, Redis, and cloud-managed messaging to process high volumes of clinical and operational events with guaranteed delivery.
- Improved system performance through query optimization, caching strategies, and indexing — reducing P95 API latency by ~50%.
- Led cloud infrastructure migration to AWS (EKS, RDS, S3, CloudFront), improving availability, deployment safety, and operational visibility.
- Integrated LLM-powered workflows to automate clinical documentation and summarization, reducing manual effort for clinicians.
- Implemented CI/CD pipelines with automated testing and blue/green deployments, enabling safe and frequent releases.
- Established observability standards (Prometheus, Grafana, centralized logging) to monitor system health and background job throughput; mentored junior engineers and drove architectural decisions.

InvoZone — Software Engineer · Backend **Dec 2020 – Dec 2022**

Pakistan

- Designed and shipped backend services for multiple cloud-native client products using Django and RESTful APIs, supporting tens of thousands of daily active users.
- Built event-driven workflows using async task processing and messaging systems to decouple request handling from long-running background jobs.
- Deployed serverless workloads on AWS Lambda and GCP Cloud Run for background processing, scheduled tasks, and API extensions.
- Containerized microservices with Docker and Kubernetes, enabling horizontal scaling and consistent environments across staging and production.
- Contributed to blockchain-integrated backend systems — implemented secure smart contract interactions and off-chain processing pipelines.
- Designed CI/CD pipelines with automated testing, reducing manual release intervention by ~80%; integrated Stripe and Auth0 for multi-tenant SaaS billing and authentication.
- Built data ingestion and transformation pipelines with a focus on reliability, idempotency, and observability.

Taleemabad — Associate Software Engineer **Jul 2019 – Dec 2020**
Pakistan

- Developed and maintained backend services for a national-scale LMS serving 100K+ active students across Pakistan.
- Built async background workflows using task queues and Redis for notifications, content sync, and analytics processing.
- Implemented real-time features using WebSockets and Redis pub/sub for live classroom interactions.
- Optimized PostgreSQL queries and data models for high-traffic endpoints; integrated search and indexing to improve content discovery across large educational datasets.
- Supported production deployments and monitoring of backend systems under real load.

OPEN SOURCE

Volta Engine — serverless microVM runtime **github.com/khanashes/volta**
Firecracker/KVM microVMs with snapshot/restore and userfaultfd lazy memory restore; resumes VMs before memory is fully copied. Targets sub-100ms cold starts, sub-200ms lazy restore. Python SDK + voltad daemon.

FlowMesh — unified event fabric **github.com/khanashes/flowmesh**
One clustered platform (Go) for key-value caching, job queues and event streaming, replacing a Redis + RabbitMQ + Kafka stack. Partitioned append-only logs with consumer groups and replay, time-travel debugging, OpenTelemetry tracing, Prometheus metrics, JSON Schema validation, Go and Python SDKs.

SKILLS

Languages	Python, SQL, Go, JavaScript, TypeScript, Bash
Backend	Django, Django REST Framework, FastAPI, Celery
Datastores	PostgreSQL, Redis, MongoDB, Elasticsearch
Cloud & infra	AWS (EC2, EKS, ECS, Lambda, RDS, S3, CloudFront), GCP (Cloud Run, Pub/Sub), Docker, Kubernetes, Helm, Terraform
DevOps & observability	GitHub Actions, Prometheus, Grafana, CloudWatch, blue/green deployments, GitOps
Security	OAuth2 / JWT, RBAC, HIPAA compliance, encryption at rest & in transit

RECOGNITION & EDUCATION

- IEEE Xtreme Programming Competition — International, Top 3 finish
- Growth Driver of the Year — InvoZone internal recognition award
- AWS Certified Cloud Practitioner (in progress) · IBM Python for Data Science · Advanced Django (Udemy)
- B.S. in Software Engineering — COMSATS Institute of Information and Technology, Islamabad (2016 – 2020)